

HTML — PAGE STRUCTURE

```
<!DOCTYPE html>

<html lang='en'>
  <head>
    <meta charset='UTF-8' />
    <meta name='viewport'
      content='width=device-width,
      initial-scale=1.0' />
    <title>My Page</title>
    <link rel='stylesheet'
      href='style.css' />
  </head>
  <body>
    <!-- visible content -->
  </body>
</html>
```

SEMANTIC TAGS

```
<header> Page / section header
<nav> Navigation links only
<main> Primary content (1 per page)
<article> Self-contained content
<section> Thematic group + heading
<aside> Sidebar / tangential
<footer> Page / section footer
<figure> Image + figcaption
<time> Dates — use datetime attr
<dialog> Native modal
<details> Native accordion — no JS
<mark> Highlighted text
```

TEXT & INLINE TAGS

```
<h1>–<h6> Headings — h1 largest
<p> Paragraph block
<strong> Bold + semantically important
<em> Italic + emphasis
<a href=''> Clickable hyperlink
<span> Inline container
<br/> Line break (self-closing)
<code> Inline code
<blockquote> Quoted block
```

LISTS, MEDIA & FORMS

```
<ul><li> Bullet list
<ol><li> Numbered list
<img src='' alt=''> Image — always add alt!
<video controls> Video player
<input type='text'> Text input
<input type='email'> Email
<input type='password'> Hidden text
<input type='checkbox'> Tick box
<select><option> Dropdown
<textarea> Multi-line text
<button type='submit'> Submit form
<label for='id'> Links to input
```

KEY ATTRIBUTES

id Unique page identifier

class Style group — reusable

href Link destination URL

src File source path / URL

alt Image description (a11y+SEO)

type Input / button type

placeholder Ghost hint text in input

required Field must not be empty

disabled Cannot interact with element

data-* Custom data attribute

aria-label Accessible name override

aria-hidden Hide from screen readers

tabindex Keyboard focus order

rel Link relationship type

CLASS vs ID

	class	id
HTML	class='card'	id='hero'
CSS	.card { }	#hero { }
Repeat?	Yes – unlimited	No – unique
Multi?	class='a b c'	One only
Use for	Styling groups	JS / anchors
Spec.	0,0,1,0	0,1,0,0

CSS SELECTORS

***** Universal — all elements

p Tag — all paragraphs

.card Class selector

#hero ID selector

.card.featured Both classes on same element

.card p p anywhere inside .card

.card > p Direct child p of .card

.card + p p immediately after .card

[type='email'] Attribute selector

:hover :focus :active — interaction states

:first-child :last-child — position in parent

:nth-child(2) Exactly the 2nd child

:nth-child(odd) 1st 3rd 5th...

:not(.skip) Excludes .skip elements

:is(h1,h2,h3) Matches any in list

:where(h1,h2) Same — zero specificity

:has(img) Has img as descendant

::before / **::after** Virtual elements — needs content:"

::placeholder Input ghost text style

::selection User highlighted text

SPECIFICITY SCORES

***** 0,0,0,0

tag 0,0,0,1

.class 0,0,1,0

#id 0,1,0,0

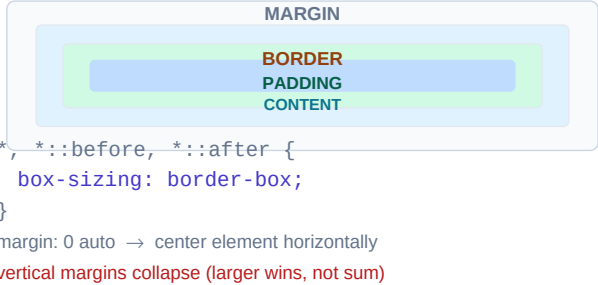
style='' 1,0,0,0

!important Override all

:where() 0,0,0,0 — use for resets

@layer later Beats all earlier layers

BOX MODEL



DISPLAY VALUES

block Full width · new line · respects w/h

inline Content width · flows in text · no w/h

inline-block Inline flow + respects w/h/margin

flex 1D layout — row or column direction

grid 2D layout — rows AND columns

none Completely removed — no space taken

display:none

Removed — no space — not interactive

visibility:hidden

Invisible — space kept — not clickable

COLORS

color:red | #FF0000 | #F00

rgb(255, 0, 0)

rgba(255, 0, 0, 0.5)

hsl(220, 90%, 56%)

background:#f8fafc | transparent

background: linear-gradient(135deg, #6366f1, #8b5cf6)

opacity:0=invisible to **1=fully visible**

TYPOGRAPHY

font-family: 'Inter', Arial, sans-serif

font-size:16px | 1rem | 1.5em

font-weight:400=normal | 700=bold

line-height:1.6 (unitless recommended)

text-align:left | center | right

text-decoration:none (removes underline)

text-transform:uppercase | capitalize

letter-spacing:0.05em

text-overflow:ellipsis + **overflow:hidden**

UNITS & SIZING

px Absolute — borders, shadows

rem Root font size (default 16px)

em Parent font size — compounds

% Relative to parent

vw / **vh** 1% of viewport width / height

ch Width of '0' — text columns

fr Grid fractional space

clamp() clamp(min, preferred, max)

BORDERS & SHADOWS

border:1px solid #e2e8f0

border-radius:8px | 50% = perfect circle

outline:2px solid #6366f1

outline-offset:2px (no layout shift)

box-shadow:0 4px 6px rgba(0,0,0,0.08)

box-shadow:inset 0 1px 3px rgba(0,0,0,0.1)

FLEXBOX — CONTAINER

disptay:flex

flex-direction:row | column | row-reverse

justify-content:flex-start | flex-end | center

space-between | space-evenly

align-items:stretch | flex-start | flex-end

center | baseline

flex-wrap:nowrap | wrap | wrap-reverse

gap:16px | row-gap col-gap

flex-flow:row wrap (direction + wrap)

align-content:center | space-between (multi-line)

FLEXBOX — ITEMS

flex:1 = 1 1 0% equal share from zero

flex:auto = 1 1 auto from content size

flex:none = 0 0 auto rigid, never changes

flex:0 0 280px fixed 280px always

flex-grow:0=stay | 1=grow into free space

flex-shrink:0=rigid | 1=shrink if needed

align-self:flex-start | center | flex-end

order:-1 before | 0 default | 1 after

Common Flex Patterns

Center anything: display:flex; place-items:center

Sticky footer: body flex-col, main flex:1

Navbar: justify:space-between, align:center

Card footer down: flex-col, footer margin-top:auto

Equal tabs: each child → flex: 1

CSS GRID — CONTAINER

display:grid

grid-template-columns:1fr 2fr 1fr

repeat(3, 1fr) | 240px 1fr

repeat(auto-fill, minmax(280px,1fr))

grid-template-rows:64px 1fr 80px

grid-template-areas:'sidebar header' 'sidebar main'

gap:24px | 16px 24px

place-items:center (align + justify)

CSS GRID — ITEMS

grid-area:sidebar (named area)

grid-column:1 / 3 | span 2 | 1 / -1

grid-row:1 / 3 | span 2

align-self:stretch | start | end | center

justify-self:stretch | start | end | center

POSITIONING

static Default — offsets do nothing

relative Offset from own position in flow

+ creates context for absolute children

absolute Removed from flow

Positioned to nearest positioned ancestor

fixed Removed — always relative to viewport

Stays put during scroll

sticky Relative until threshold → then fixed

Needs top / left / right / bottom!

z-index: Only works on positioned elements

inset:0 = top + right + bottom + left: 0

PSEUDO-CLASSES & ELEMENTS

:hover / **:focus** / **:active** Interaction states

:disabled / **:checked** Form states

:valid / **:invalid** Form validation states

:first-child / **:last-child** Position in parent

:nth-child(n/odd/even) Pattern selection

:not(.cls) / **:has(img)** Exclude / parent selector

::before / **::after** Virtual — needs content:"

::placeholder Input ghost text style

CSS CUSTOM PROPERTIES

```
:root {
  --color-primary: #6366f1;
  --color-bg: #ffffff;
  --color-text: #0f172a;
  --space-4: 1rem;
  --radius: 8px;
}
.card {
  color: var(--color-text);
  padding: var(--space-4);
}
color: var(--x, #6366f1); /* fallback */
[data-theme='dark'] {
  --color-bg: #0f172a;
}
document.documentElement
  .dataset.theme = 'dark';
```

TRANSITIONS & ANIMATIONS

```
/* Transition */
transition: transform 200ms ease,
           opacity 200ms ease;

/* Keyframes */
@keyframes fadeSlideIn {
  from { opacity: 0;
    transform: translateY(20px); }
  to { opacity: 1;
    transform: translateY(0); }
}
animation: fadeSlideIn 600ms ease forwards;
animation-delay: calc(var(--i) * 80ms);
animation-iteration-count: infinite;
@media (prefers-reduced-motion: reduce) {
  * { animation-duration: 0.01ms !important; }
```

ANIMATE ONLY — PERFORMANCE

transform GPU composited — no reflow or repaint

opacity GPU composited — no reflow or repaint

color / **bg** Repaint only — acceptable

width / **height** Reflow! Use transform: scale() instead

top / **left** Reflow! Use transform: translate() instead

margin / **pad** Reflow! Never animate these

will-change: transform — use sparingly!

contain: layout style — isolate paint scope

font-display: swap — prevent invisible text

RESPONSIVE — MEDIA QUERIES

```
/* Mobile-first: base = mobile */
.grid { grid-template-columns: 1fr; }
@media (min-width: 640px) {
  .grid { grid-template-columns:
    repeat(2, 1fr); }
}
@media (min-width: 1024px) {
  .grid { grid-template-columns:
    repeat(3, 1fr); }
}
/* Fluid — no media queries needed */
repeat(auto-fill, minmax(280px,1fr));
font-size: clamp(1rem, 5vw, 3rem);
@media (prefers-color-scheme: dark)
@media print { .nav { display:none } }
```

CONTAINER QUERIES

```
/* Parent: define context */
```